Introduction to Software Engineering

COSC 336

ReSource (Phase III)

Requirements Document

Sections prepared by Rashid Alzaabi

Intelligent AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System

Dr. Ala' Alsamarneh
Dr. Hamdan Alshkeili
Dr. Natnael Takele
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Prepared by:
Rashid Alzaabi 100064869
Prepared for:
Head of Facilities Management, Khalifa University

Proposed Technical Owner:
IT Department, Khalifa University

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2. System Features

2.5 Maintenance and Asset Life-Cycle Management (FR-5)

2.5.1 Description and Priority

This feature lets authorized users report defects, request inspections, assign and record repair and refurbishment work and its cost, and update an asset's condition after service. It keeps a complete, unchangeable life-cycle history for every asset, from acquisition to final disposition, and manages end-of-life actions (donation, recycling, disassembly for usable components, retirement and disposal) through a safety review and an approval, with extra safeguards for hazardous items. It refines Phase 2 requirement FR-5 and the project's "Inspection and Asset Life-Cycle Tracking" functionality.
Priority: High. This feature develops Phase 2 requirement FR-5. Maintenance and repair tracking is part of the minimum prototype [1], and the life-cycle history is needed by the sustainability indicators (FR-13).

2.5.2 Stimulus/Response Sequences

1. Defect reported: an Asset Coordinator, Department Representative or Maintenance Staff member reports a defect → the system creates a maintenance request in the "Reported" state and notifies Maintenance Staff.
2. Work assigned and recorded: a request is assigned to a Maintenance Staff member, who starts the inspection and records findings and repair work → the system moves the request through its states, saves the work and cost, updates the asset's condition and availability, and adds life-cycle events.
3. End-of-life proposed: an Asset Coordinator or Sustainability Officer proposes donation, recycling, disassembly, retirement or disposal → the system checks for blocking items, routes the proposal to a safety review by a different Sustainability Officer and then to approval, and on completion sets the asset's final state and updates the sustainability indicators.
4. History viewed: an authorized user opens an asset's history → the system lists every life-cycle event in date order.

2.5.3 Functional Requirements

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
A. Maintenance and repair				
FR-5.1	The system shall allow an Asset Coordinator, Department Representative or Maintenance Staff member to report a defect in a registered asset, recording the asset, defect description, date reported, reporting user and optional photographs.	Essential	UC-6	A report with all required fields creates a maintenance request in the 'Reported' state, linked to the asset and shown in its maintenance list.
FR-5.2	The system shall allow an Asset Coordinator or Maintenance Staff member to request an inspection of an asset, stating the reason and the requested inspection date.	Essential	UC-6	The request appears in the Maintenance Staff queue in the 'Inspection Requested' state.
FR-5.3	The system shall allow Maintenance Staff or an Administrator to assign a maintenance request to a Maintenance Staff member, who is then notified.	Essential	UC-6	The assigned request appears in that member's list, and the member receives a notification.
FR-5.4	The system shall move a maintenance request to 'Under Inspection' when the assigned Maintenance Staff member starts the inspection, and to 'Repair in Progress' when they start repair or refurbishment work.	Essential	UC-6	Each start action changes the state as stated and records the date, time and user.
FR-5.5	The system shall allow Maintenance Staff to record inspection results: inspection date, inspector, findings, assessed condition, whether the asset is safe to use, and a recommended next action (no action, repair, refurbish or end-of-life review).	Essential	UC-6	All listed fields are saved and appear in the asset's maintenance and life-cycle history.
FR-5.6	The system shall allow Maintenance Staff to record repair and refurbishment work: work type (repair or refurbishment), work performed, parts replaced, start and completion dates, labor hours, cost in AED, and the estimated additional years of use the work adds.	Essential	UC-6	A saved record shows all fields, and its cost is added to the asset's cumulative maintenance cost.

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-5.7	The system shall track each maintenance request through the maintenance states defined in Appendix B (Reported, Inspection Requested, Under Inspection, Repair in Progress, Completed, Closed – Not Repairable, Cancelled) and show the current state to the reporting user and to the asset's Asset Coordinator.	Essential	UC-6	Each state change is visible to both users and is recorded with date, time and user.
FR-5.8	When a maintenance request is completed, the system shall require Maintenance Staff to record the asset's updated condition, and shall update the asset's condition and availability in its record.	Essential	UC-6	A request cannot be marked 'Completed' without an updated condition; the asset record shows the new condition immediately afterwards.
FR-5.9	While an asset has a maintenance request in the 'Under Inspection' or 'Repair in Progress' state, the system shall not show it as available in search, offer it in matching results, or allow it to be reserved or transferred.	Essential	UC-2, UC-3, UC-5, UC-6, UC-9	The asset does not appear as available in search or matching, and reserve or transfer attempts are rejected with a message naming the open maintenance request.
FR-5.10	The system shall display each asset's cumulative maintenance cost (repairs and refurbishments) and number of maintenance actions to the roles allowed to see financial information (SEC-5).	Essential	UC-6	The displayed total equals the sum of the recorded costs for that asset.
FR-5.11	When an asset’s cumulative maintenance cost reaches a configurable percentage of its replacement price (default 50%, TBD-7), the system shall flag the asset for end-of-life review and notify its Asset Coordinator and the Sustainability Officer.	Desirable	UC-6, UC-7	Recording work that crosses the threshold sets the flag and sends both notifications.
FR-5.12	When Maintenance Staff close a request as 'Closed – Not Repairable', the system shall flag the asset for end-of-life review and notify its Asset Coordinator and the Sustainability Officer.	Essential	UC-6, UC-7	Closing a test request as not repairable sets the flag and sends both notifications.
FR-5.13	The system shall allow the user who reported the defect, the asset's Asset Coordinator or an Administrator to cancel a maintenance request that is not yet completed, with a reason. Cancelled requests remain in the history.	Essential	UC-6	A cancelled request shows the 'Cancelled' state, the reason and the user; it cannot be cancelled without a reason.

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-5.14	The system shall reject a maintenance record that is missing a required field, has a completion date before its start date, has a work or completion date in the future, or has a negative cost, and shall show which field is invalid.	Essential	UC-6	Each invalid case is rejected, nothing is saved, and the invalid field is identified.
FR-5.15	The system shall not allow users without the Maintenance Staff or Administrator role to record inspection or repair results.	Essential	UC-6	Attempts by any other role are denied and written to the audit log.
B. Life-cycle history				
FR-5.16	The system shall automatically record a life-cycle event for acquisition (registration), allocation, change in utilization status, inspection, maintenance, repair, transfer, refurbishment, donation, recycling, disassembly, retirement and disposal. Each event shall include date and time, event type, the user who performed it (by internal user ID), the user who approved it (if any) and a reference to the related record.	Essential	UC-1, UC-4, UC-5, UC-6, UC-7	Performing each event type in a test adds exactly one correctly typed event to the asset's history.
FR-5.17	The system shall display an asset's life-cycle history in date order, with filters for event type and date range, to the asset's Asset Coordinator, Maintenance Staff, Sustainability Officers, Administrators and members of the department holding the asset.	Essential	UC-6, UC-7	The listed roles see all events oldest to newest; each filter returns only matching events; other users cannot open the history.
FR-5.18	The system shall not allow any user, including Administrators and System Administrators, to edit or delete a recorded life-cycle event. A correction shall be recorded as a new event that references the original.	Essential	UC-6, UC-7	No edit or delete option exists; a correction appears as a new linked event and the original is unchanged.
FR-5.19	The system shall show each asset's age (time since its purchase date) and time in service (time since it was first allocated for use).	Desirable	UC-2, UC-7	Both displayed values equal the time between today and the recorded purchase and first-allocation dates.
C. End-of-life actions: donation, recycling, disassembly, retirement and disposal				
FR-5.20	The system shall allow an Asset Coordinator or Sustainability Officer to propose an end-of-life action for an asset (donation, recycling, disassembly for usable components, retirement or disposal) with a written justification.	Essential	UC-7	A proposal without a justification cannot be submitted; a valid proposal is saved in the 'Pending Safety Review' state.

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-5.21	The system shall not accept an end-of-life proposal for an asset that has an open reservation, a pending transfer or an open maintenance request, and shall list the blocking items.	Essential	UC-7	A proposal for such an asset is rejected and the blocking items are listed.
FR-5.22	When an end-of-life action is proposed, the system shall show the open requests the asset may satisfy (same category, and requested quantity not more than the asset's available quantity) and, when available, the sustainable-action recommendation from FR-10 with its confidence and explanation.	Desirable	UC-7	For a test asset that matches an open request by category and quantity, the request is shown on the proposal screen.
FR-5.23	The system shall show the hazard flag recorded for the asset at registration (FR-1) and require the proposer to confirm or update it. A Sustainability Officer other than the proposer shall then record a safety review confirming or correcting the classification; until then the proposal stays in 'Pending Safety Review'.	Essential	UC-7	A proposal cannot leave 'Pending Safety Review' until a different Sustainability Officer records the review.
FR-5.24	For assets confirmed as hazardous or electronic waste, the safety review shall assign a handling party (Facilities Management or a vendor on the approved list in FR-5.25), and the system shall not allow the action to be completed without the handling party's record and certificate.	Essential	UC-7	A hazardous asset cannot pass the safety review without a handling party, or be completed without the handling record and certificate.
FR-5.25	The system shall allow an Administrator to maintain the list of approved recycling and disposal vendors (name, materials handled, contact details and approval expiry date). Only Facilities Management or a vendor on the list with an unexpired approval can be assigned as a handling party.	Essential	UC-7	A vendor not on the list, or with an expired approval, cannot be assigned.
FR-5.26	After the safety review, the system shall set the proposal to ‘Pending End-of-Life Approval’, route it through the approval stages in BR-8, and shall not change the asset to its final state until all required approvals and the completion record (FR-5.29, FR-5.30 or FR-5.31) are recorded.	Essential	UC-7	The asset reaches its final state only after the last approval and a complete completion record.

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-5.27	While an asset is 'Pending Safety Review' or 'Pending End-of-Life Approval', the system shall not show it as available in search, offer it in matching results, or allow it to be reserved or transferred.	Essential	UC-2, UC-3, UC-5, UC-7, UC-9	Search, matching, reserve and transfer attempts on a pending test asset do not succeed.
FR-5.28	When an end-of-life proposal is rejected at safety review or approval, the system shall record the reason and return the asset to its previous state.	Essential	UC-7	After rejection the asset is back in its previous state and the reason appears in its history.
FR-5.29	When a donation is completed, the system shall record the recipient organization, date, quantity, weight in kilograms and handover document.	Essential	UC-7	A completed donation shows all fields; a missing required field blocks completion.
FR-5.30	When recycling, disassembly or disposal is completed, the system shall record the method, handling party, date, quantity, weight in kilograms and the recycling or disposal certificate (required for hazardous and electronic-waste items; recorded when issued for other items).	Essential	UC-7	The completed record shows all fields; a hazardous item cannot be completed without its certificate.
FR-5.31	When a retirement is completed, the system shall record the date, reason, final location (for example central storage) and the user who completed it.	Essential	UC-7	A completed retirement shows all fields; a missing field blocks completion.
FR-5.32	Weight shall be recorded for every completed donation, recycling, disassembly and disposal: the actual weight if known, otherwise the category default weight; if no default exists, the user shall enter an estimated weight. The record shall show whether the weight is actual, a default or an estimate.	Essential	UC-7	No completion record can be saved without a weight and its type.
FR-5.33	When a disassembly is completed, the system shall allow usable components to be registered as new asset records linked to the original asset.	Future	UC-1, UC-7	A new component record links to its parent asset, and the parent shows the link.
FR-5.34	The system shall notify the proposer, the asset's Asset Coordinator and the user whose action is needed next (safety reviewer or approver) when an end-of-life proposal is submitted, passes or fails the safety review, is approved or rejected, and when it is completed, using the notifications defined in FR-7.	Essential	UC-7	Each listed event produces notifications to the listed users.

2.7 Notifications, Reports and Audit Log (FR-7)

2.7.1 Description and Priority

This feature informs users of workflow events that need their attention, provides conventional, rule-based operational reports with fixed totals and tables, and keeps a complete, tamper-proof audit log of user and system actions. It refines Phase 2 requirement FR-7 and the project's requirements for notifications, operational reports and auditability. Notifications are in-application messages; email copies can be added once the delivery service is confirmed (TBD-6). Unlike the AI-generated reports in FR-12, the operational reports are deterministic: the same data and filters always give the same result, which makes them the reference source for every figure the system shows.

Priority: High. This feature develops Phase 2 requirement FR-7. Notifications and basic operational reports are part of the minimum prototype [1], and the audit log is required for security and legal compliance (3.3, 4.2).

2.7.2 Stimulus/Response Sequences

1. Workflow event recorded: a request, approval, transfer, maintenance, end-of-life or account event is saved → the system creates in-application notifications for the recipients in Table 2.7-1 and retries any failed delivery.
2. Report requested: an authorized user selects a report type and filters → the system generates the report from current data, limited to what the user may see.
3. Any recorded action**:** a user creates, changes, approves or rejects a record, signs in, is denied access, or uses an AI output → the system writes an audit entry automatically.
4. Audit search: a System Administrator or Administrator searches the audit log → the system lists the matching entries.

2.7.3 Functional Requirements

Report and audit requirements (FR-7.R, FR-7.A)

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
A. Standard operational reports				

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-7.R1	The system shall allow Asset Coordinators, Department Representatives, Procurement Team members, Finance Officers, Sustainability Officers, Administrators and University Management to generate these standard reports: available assets; open requests; completed transfers; maintenance history; ownership and custody changes; approval decisions; asset status by life-cycle state; and end-of-life actions (donations, recycling, disassembly, retirement, disposal).	Essential	UC-12	Each listed role can generate each of the eight reports from test data; Maintenance Staff and System Administrator accounts cannot.
FR-7.R2	The system shall allow each standard report to be filtered by date range, department, category, location and status.	Essential	UC-12	Each filter returns only matching records.
FR-7.R3	A report shall include only records from the user's own department, plus published surplus assets, unless the user's role covers all departments (Procurement Team, Finance Officer, Sustainability Officer, Administrator, University Management). Financial fields (estimated values, replacement prices, costs, savings) shall be shown only to roles allowed to see financial information (SEC-5).	Essential	UC-12	A Department Representative's report contains only their department's records and published assets, and no financial fields.
FR-7.R4	Each report shall show its title, the filters applied, the date and time generated, the generating user and the number of records.	Essential	UC-12	All five items appear on every report.
FR-7.R5	Standard reports shall be calculated directly from recorded data without AI, so the same data and filters always produce the same results.	Essential	UC-12	Generating a report twice with unchanged data gives identical results.
FR-7.R6	When no records match the selected filters, the system shall show an empty report stating that no records matched, instead of an error.	Essential	UC-12	Filters with no matches return the message and no error.
FR-7.R7	The system shall allow standard reports to be downloaded as a PDF file and as a spreadsheet-compatible (CSV) file.	Desirable	UC-12	Both files open and contain the same records as the on-screen report.
FR-7.R8	The system shall allow a Finance Officer, Sustainability Officer or Administrator to schedule a standard report to be generated monthly and delivered by notification.	Future	UC-12	A scheduled report is generated on the first day of each month and the user is notified.

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
B. Audit log				
FR-7.A1	The system shall record an audit entry for every create, update, approval, rejection, cancellation, AI-recommendation override, sign-in, failed sign-in, denied access attempt, role or permission change, consent given or withdrawn, human-review request, data export, and view of asset or transfer records. Each entry shall include date and time, internal user ID, role, action, affected record, and the old and new values of changed fields; entries shall never store passwords or personal-data field values.	Essential	UC-1 to UC-13	Each listed action in a test produces one complete audit entry, and no entry contains a password, name, university ID or email.
FR-7.A2	The system shall not allow any user, including System Administrators, to edit or delete audit entries.	Essential	UC-1 to UC-13	No edit or delete function exists, and attempts through the interface are denied and logged.
FR-7.A3	The system shall allow only System Administrators and Administrators to view the audit log, and to search it by user, record, action type and date range.	Essential	UC-8	Other roles cannot open the audit log; each search filter returns only matching entries.
FR-7.A4	For each AI output used in a decision (classification, match, recommendation, assistant action or report), the system shall record the model version, the records used as input, the output, its confidence, its explanation, and whether the user accepted, edited or rejected it (with the user's reason for an override).	Essential	UC-7, UC-9, UC-10, UC-11, UC-12	Every AI-assisted decision in a test has a complete entry, including model version and explanation.
FR-7.A5	The system shall keep audit entries for at least the retention period set by Khalifa University IT policy (TBD-4); until that period is confirmed, no audit entry shall be deleted.	Essential	UC-1 to UC-13	No audit entries are removed during the test period.
FR-7.A6	If an audit entry cannot be recorded, the system shall not complete the action and shall show an error.	Essential	UC-1 to UC-13	With audit recording forced to fail, the action is not saved and an error is shown.

2.12 AI-Generated Reports (FR-12)

2.12.1 Description and Priority

This feature turns recorded operational and sustainability data into written reports: monthly summaries, savings and sustainability reports, sustainability updates, management bulletins, high-demand-category summaries, maintenance insights and lists of recommended actions. It refines Phase 2 requirement FR-12 and the project's Generative AI Reporting functionality. Where a classic report shows fixed totals and tables (FR-7), an AI report can summarize trends, explain unusual changes and suggest management actions. Because AI text can be wrong, every figure in an AI report must come from the deterministic reports and indicators (FR-7, FR-13), assumptions and missing data must be stated, and a person must approve the report before anyone else can see it.

Priority: High for the monthly activity summary and the savings and sustainability report (the minimum prototype includes basic AI-generated reports [1]); Medium for the other report types. This feature develops Phase 2 requirement FR-12.

2.12.2 Stimulus/Response Sequences

1. Report requested: an authorized user selects an AI report type, period and scope → the system calculates the underlying figures, produces a draft labeled as AI-generated, and checks its figures.
2. Draft reviewed: the user edits and approves the draft → the system re-checks the figures, stores both versions, records the approval and makes the report available to authorized recipients.
3. AI unavailable: the AI service fails or times out → the system informs the user and offers the standard reports and indicators instead.

2.12.3 Functional Requirements

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-12.1	The system shall allow a Finance Officer, Sustainability Officer, Procurement Team member or Administrator to request an AI-generated report for a selected period and scope (campus or a department).	Essential	UC-12	Each of the four roles can request a report; other roles cannot.

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-12.2	The system shall offer these AI report types, each containing at least the listed content: (a) monthly activity summary – counts of new listings, requests, transfers, maintenance actions and end-of-life actions compared with the previous month; (b) savings and sustainability report – indicators 1–9 (Table 2.13-1) with their assumptions; (c) sustainability update – reuse, waste diversion and carbon indicators; (d) management bulletin – the three largest changes and recommended actions; (e) high-demand-category summary – the five most requested categories and unmet requests; (f) maintenance insights – most-repaired categories, maintenance cost and not-repairable rate; (g) recommended actions list.	Essential (a, b); Desirable (c–g)	UC-12	Each draft contains every item listed for its type.
FR-12.3	The system shall base each AI report only on data recorded in the system for the selected period and within the requesting user's permitted scope.	Essential	UC-12	A department-scope report contains no data from other departments.
FR-12.4	Every figure stated in an AI report shall equal the corresponding value in the standard reports (FR-7) or indicators (FR-13), or a change calculated from them, for the same period. The system shall check the draft and again check the version submitted for approval, flag any statement whose figure does not match, and not allow the report to be approved until it is corrected.	Essential	UC-12	100% of figures in approved test reports match the calculated values; a wrong figure added during editing is flagged and blocks approval.
FR-12.5	Each AI report shall list the assumptions and missing data behind any estimate, and shall label sustainability and savings figures as estimates.	Essential	UC-12	Every report has an 'Assumptions and missing data' section, and every estimated figure carries the label.
FR-12.6	Each AI report shall show an 'AI-generated' label, the generation date, the period covered, the model version and the data sources used.	Essential	UC-12	All five items appear on every AI report.
FR-12.7	Each recommended action in an AI report shall include a short explanation and a confidence level (AI-10). The system shall not carry out any recommended action automatically.	Essential	UC-12	Every recommendation shows both items, and no record changes as a result of generating a report.

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-12.8	The system shall highlight changes larger than a configurable percentage (default 25%, TBD-13) compared with the previous period, calculated from FR-7 and FR-13 values, and give a possible explanation for each.	Desirable	UC-12	A 30% change in test data is highlighted with an explanation; a 10% change is not.
FR-12.9	An AI report shall remain a ‘Draft’ until an authorized reviewer reviews it, edits it if needed and approves it. A report that contains financial figures shall be approved only by a Finance Officer or an Administrator; a report without financial figures may also be approved by a Sustainability Officer (BR-11). Only approved reports can be shared with University Management or other users, and financial figures in a shared report shall be hidden from recipients not allowed to see financial information (SEC-5).	Essential	UC-12	Recipients cannot open a draft; after approval they can, and a recipient without financial access sees no financial figures. A Sustainability Officer cannot approve a report that contains financial figures.
FR-12.10	The system shall keep both the original AI draft and the approved version of each report, and record who edited and approved it. Discarded drafts shall be kept with the state 'Discarded'.	Essential	UC-12	Both versions and the approver can be retrieved for every published report; discarded drafts are retrievable.
FR-12.11	If the AI service is unavailable, fails or does not return a usable result within the time limit in PER-5, the system shall inform the user and offer the standard reports (FR-7) and indicators (FR-13) for the same period instead.	Essential	UC-12	With the AI service disabled, the user sees the message and can open the standard report.
FR-12.12	If the selected period has no recorded activity, the system shall say so and shall not generate a written summary.	Essential	UC-12	An empty period returns the message and no narrative text.

2.13 Sustainability Indicators and Dashboard (FR-13)

2.13.1 Description and Priority

This new feature calculates the sustainability indicators the project asks for (waste diversion, avoided purchases, asset-life extension, reuse rate, repair, financial savings and estimated carbon-emission reductions) and shows them on a dashboard. The indicators are calculated with fixed, published formulas rather than by AI, so they can be checked by hand and validated during system testing.

Each indicator shows its formula, its assumptions and how much of it relies on default values, which is how the system communicates the uncertainty in its sustainability estimates. The

figures are estimates for internal planning; consistent with the Phase 2 scope, they are not certified carbon or financial accounts.

Priority: High. Sustainability indicators and dashboard summaries are part of the minimum prototype [1].

2.13.2 Stimulus/Response Sequences

1. Action completed: a transfer, repair, refurbishment, donation, recycling, disassembly, retirement or disposal is completed → the system includes it in the indicators for that period.
2. Dashboard opened: a user selects a period and scope → the system shows each indicator the user may see, its change from the previous period, its trend and its data quality.
3. Value updated: a Finance Officer or Sustainability Officer updates a category default value → the system saves it as a new version with its source and effective date.

2.13.3 Indicator Definitions

Table 2.13-1: Sustainability indicator definitions

#	Indicator	How it is calculated	Unit
1	Assets reused	Number of completed internal transfers that fulfilled a departmental request, counted per unit of quantity.	count
2	Reuse rate	Assets reused ÷ (assets reused + donated + recycled + disassembled + retired + disposed) × 100. Shows how many of the assets leaving a department stayed in use on campus.	%
3	Avoided purchases	Count = assets reused (indicator 1). Value = Σ (quantity × replacement price), using the replacement price recorded on the request or asset, or the category default.	count, AED
4	Financial savings (net)	Avoided purchase value + avoided storage and handling cost (per reused asset, category default) - transfer costs recorded on the completed transfers - maintenance costs recorded on the reused assets between their publication as surplus and their transfer.	AED
5	Waste diverted	Total weight of assets reused, donated, recycled or disassembled (FR-5.32 weight rule). Each asset unit is counted once per period, even if it was refurbished before being reused.	kg
6	Waste diversion rate	Waste diverted ÷ (waste diverted + weight of assets disposed) × 100.	%

#	Indicator	How it is calculated	Unit
7	Repairs	Number of maintenance requests completed with work type repair or refurbishment (FR-5.6). Repair success rate = those requests ÷ (those requests + requests closed as not repairable) × 100.	count, %
8	Asset-life extension	Σ additional years of use for repaired, refurbished or reused assets: the Maintenance Staff estimate recorded under FR-5.6 where present, otherwise the category default. Each asset unit is counted once per period.	years
9	Estimated carbon reduction	Σ (quantity of reused assets × the category emission factor for producing a new item). Each asset unit is counted once per period.	kg CO_2e

2.13.4 Functional Requirements

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-13.1	The system shall calculate the nine indicators in Table 2.13-1, using the formulas shown, for any selected period, department and category.	Essential	UC-12	For a prepared test data set, every indicator equals a hand calculation using the same formulas and default values.
FR-13.2	The system shall calculate indicators with fixed formulas, not with AI, so the same data always gives the same values.	Essential	UC-12	Recalculating with unchanged data gives identical values.
FR-13.3	The system shall count only completed actions. When a completed action is corrected by a linked correction record (SEC-12), the indicators shall use the corrected values, and no asset unit shall be counted twice for the same action.	Essential	UC-5, UC-6, UC-7, UC-12	Recording a correction to a completed test transfer updates every affected indicator, and the original is not counted twice.
FR-13.4	The system shall allow a Finance Officer to maintain replacement prices and storage and handling costs, and a Sustainability Officer to maintain default weights, additional years of use and emission factors, for each asset category. Every value shall have a source and an effective date.	Essential	UC-12	A value cannot be saved without a source and an effective date.
FR-13.5	The system shall keep every version of each category value and record which version was used in each calculation, so past results can be reproduced.	Essential	UC-12	After a value changes, a past period's result is reproduced exactly using the old version.
FR-13.6	When an actual value (such as price or weight) is recorded for an asset, the system shall use it; otherwise it shall use the category default and mark the record as using a default.	Essential	UC-1, UC-7, UC-12	Records with and without actual values use the correct value and are marked correctly.

ID	Requirement	Priority	Use case	Acceptance criterion (done when…)
FR-13.7	The system shall label an indicator 'Estimate' when any contributing record uses a default or estimated value, and shall show the percentage of contributing records that did.	Essential	UC-12	An indicator with 3 of 10 records on defaults shows 'Estimate – 30% default values'.
FR-13.8	If a value an indicator needs (such as replacement price, emission factor or additional years of use) has neither an actual value nor a category default for a record, the system shall exclude the record from that indicator and list it under 'Missing data' with a count.	Essential	UC-12	A test record with no price and no default is excluded from indicator 3 and listed.
FR-13.9	The dashboard shall show each indicator for the selected period, its change from the previous period, and its trend over the last six months (or since the system started being used, if shorter).	Essential	UC-12	Displayed values and changes match the calculated results.
FR-13.10	When a user selects an indicator, the dashboard shall show its definition, formula and assumptions.	Essential	UC-12	Selecting any indicator shows all three.
FR-13.11	The dashboard shall show the non-financial indicators (1, 2, 5, 6, 7, 8, 9 and the count part of 3) to all signed-in users, for the whole campus and for their own department. The financial indicators (the AED value in 3, and 4) shall be shown only to the roles allowed to see financial information in SEC-5.	Essential	UC-12	A Department Representative sees the non-financial indicators for the campus and their own department, and no AED values.
FR-13.12	The dashboard shall state that its figures are estimates for internal planning and are not certified carbon or financial accounts.	Essential	UC-12	The statement is visible on the dashboard.
FR-13.13	The system shall allow indicator results to be downloaded as a CSV file that lists the category value versions used, containing only the indicators the user may see.	Desirable	UC-12	The downloaded file matches the dashboard and lists the versions used.

3. Other Nonfunctional Requirements

3.2 Safety Requirements

These requirements prevent loss, damage or harm that could result from using the system, especially where AI recommendations could influence transfer, repair, donation, recycling or disposal decisions. They keep high-impact decisions under human control, separate the person who proposes an action from the people who check and approve it, and close the safety gaps identified in Phase 2 for hazardous and risky items. External procedures that apply: Khalifa University Facilities Management procedures and approved e-waste and recycling vendors for hazardous items (to be confirmed, TBD-7).

ID	Requirement	How it is measured	Use case	Priority
SAF-1	Assets flagged as hazardous or electronic waste shall not be matched, reserved, transferred, donated, recycled or disposed of until a Sustainability Officer has recorded a safety review; their final disposition also requires an assigned handling party (Facilities Management or an approved vendor, FR-5.25).	Test: 100% of match, reserve, transfer and end-of-life completion attempts on an unreviewed hazardous test asset are blocked.	UC-1, UC-2, UC-3, UC-5, UC-7, UC-9	Essential
SAF-2	No AI output shall automatically carry out a transfer, repair, donation, recycling, retirement or disposal; each is carried out only by an explicit action of an authorized user (an approval, or Maintenance Staff recording work).	Test: 0 record changes caused directly by AI outputs across all AI test cases; the audit log shows the user who performed every such action.	UC-4 to UC-7, UC-9 to UC-12	Essential
SAF-3	The user who proposes an end-of-life action shall not perform its safety review or approve it.	Test: 100% of safety-review and approval attempts by the proposer are rejected.	UC-7	Essential
SAF-4	Disposal shall be the last option: a disposal proposal shall require a separate reason for each alternative (reuse, repair, donation and recycling) explaining why it is not suitable.	Test: in 100% of attempts, a disposal proposal with any of the four reasons left empty cannot be submitted.	UC-7	Essential
SAF-5	An asset recorded as not safe to use after inspection shall not be published, matched, reserved, transferred or donated until Maintenance Staff record an inspection marking it safe. Existing listings shall be withdrawn, and Department Representatives with open or approved requests for the asset shall be notified (Table 2.7-1).	Test: 100% of publish, match, reserve, transfer and donate attempts on an unsafe test asset are blocked; its listing disappears and the requesting Department Representative is notified.	UC-1, UC-2, UC-3, UC-5, UC-6	Essential

ID	Requirement	How it is measured	Use case	Priority
SAF-6	When an AI recommendation for recycling, retirement or disposal has Low confidence, the system shall mark it clearly and require the approver to record their own reason before approving.	Test: approving a Low-confidence disposal recommendation without a reason is blocked in 100% of attempts.	UC-7	Essential
SAF-7	When an asset with recorded handling or safety notes (for example heavy, fragile or containing batteries) is transferred or donated, the system shall show the notes to the receiving party before handover.	Test: the notes appear on the handover screen for 100% of test assets that have notes.	UC-5, UC-7	Desirable
SAF-8	The system shall back up all records at least once every 24 hours, so that no more than 24 hours of data can be lost. Restoration shall meet the 60-minute recovery target in QUA-6.	Inspection of the backup log shows a completed backup in every 24-hour period of the test. The QUA-6 restore test loses no more than 24 hours of changes.	UC-1 to UC-13	Essential
SAF-9	A transfer of a risky asset (hazardous, in Poor condition, or with safety notes) shall not be completed unless an inspection marking it safe was recorded within the last 90 days (configurable, TBD-7).	Test: completing a transfer of a risky test asset with no inspection in the last 90 days is blocked in 100% of attempts.	UC-5, UC-6	Essential

3.3 Security Requirements

These requirements cover authentication, authorization, data minimization, secure storage, controlled access to financial and operational information, audit trails and protection against unauthorized changes, as required by the project description [1]. They also turn the Phase 2 legal findings (the UAE Personal Data Protection Law, Federal Decree-Law No. 45 of 2021 [6], [9], [10], and the UAE Cybercrime Law, Federal Decree-Law No. 34 of 2021 [8]) and the Phase 2 AI risks into checkable requirements. Account creation, sign-in and roles are specified functionally in FR-6; the requirements below set the security rules those functions must meet.

ID	Requirement	How it is measured	Use case	Priority
SEC-1	All functions and data shall require the user to be signed in, except the sign-in, account registration, consent and password-reset pages.	Test: 100% of other pages and functions accessed without signing in are refused.	UC-1 to UC-13	Essential
SEC-2	Passwords shall have at least 8 characters, including letters and numbers, unless Khalifa University IT policy sets stricter rules (TBD-3).	Test: every password that breaks the rule is rejected.	UC-13	Essential

ID	Requirement	How it is measured	Use case	Priority
SEC-3	An account shall be locked for 15 minutes after 5 consecutive failed sign-in attempts. The sign-in screen shall state that the account is temporarily locked, and an in-application notification shall be sent to the account holder; an email copy shall be added once the email service is confirmed (TBD-6).	Test: a sign-in attempt during the lock period is refused even with the correct password, the lock message is shown and the notification is recorded.	UC-1 to UC-13	Essential
SEC-4	A signed-in session shall end after 30 minutes without activity.	Test: after 30 minutes idle, the next action requires signing in again.	UC-1 to UC-13	Essential
SEC-5	Each action shall be allowed only if the user's role and department permit it (Sections 5.1 and 3.5). Financial information (estimated values, replacement prices, costs and savings) shall be visible only to: Finance Officers, Procurement Team members, Administrators and University Management (all records); Asset Coordinators (assets they hold); Maintenance Staff (maintenance costs of assets they work on); and Sustainability Officers (aggregated savings indicators only). Filtering by estimated value shall be available only to these roles.	Permission test matrix covering every role: 100% of unauthorized attempts are denied and logged; no financial field is visible outside the listed access.	UC-2, UC-6, UC-7, UC-8, UC-9, UC-12	Essential
SEC-6	Passwords shall be stored only in a protected, non-reversible form, and personal and financial data shall be encrypted when stored and when transmitted.	Inspection of stored user records: 0 readable passwords. Test: every page and data request uses an encrypted connection, and 100% of unencrypted connection attempts are refused.	UC-8, UC-13	Essential
SEC-7	The system shall collect only the personal data it needs to operate: name, university ID, university email, department and role.	Review of the data requirements (4.1): no other personal fields are stored.	UC-8, UC-13	Essential

ID	Requirement	How it is measured	Use case	Priority
SEC-8	Before storing a user's data, the system shall obtain and record explicit consent, including the date and the version of the terms accepted (UAE PDPL [6], Article 6); for invited accounts, consent is recorded before the account is activated (FR-6.4). Audit entries and life-cycle history identify users only by an internal user ID. If a user withdraws consent, the account shall be deactivated at once, and the user's name, university ID and email shall be removed from their user record within the period approved under TBD-4 (proposed prototype default: 24 hours), leaving locked records unchanged.	Test: no account can be used without consent; 24 hours after withdrawal, the user's name, university ID and email appear nowhere in the system, and audit entries remain intact.	UC-8, UC-13	Essential
SEC-9	Users shall be able to view the personal data the system holds about them and request corrections.	Test: a user can view their data, and a correction request reaches a System Administrator.	UC-8	Essential
SEC-10	Any user affected by an AI match or recommendation shall be able to request a human review of it. The request shall go to an Administrator and be recorded.	Test: a review request reaches an Administrator and appears in the audit log.	UC-7, UC-9	Essential
SEC-11	Only the asset and request fields needed for an AI task shall be sent to AI services; user names, university IDs and email addresses shall never be sent.	Inspection of AI request logs during testing: 0 personal fields found.	UC-9, UC-10, UC-11, UC-12	Essential
SEC-12	Approved and completed records (approvals, completed transfers, end-of-life records, life-cycle events and audit entries) shall be locked against changes; corrections shall be new, linked records.	Test: 100% of edit attempts on locked records are refused.	UC-4, UC-5, UC-6, UC-7	Essential
SEC-13	The system shall accept only image and PDF uploads of up to 10 MB, and shall treat all user input as data and never execute it (for example as a script or database command).	System security test against the OWASP Top 10 checklist [13]: 0 high or critical findings; 100% of disallowed file types and files over 10 MB are rejected.	UC-1, UC-5, UC-6, UC-10	Essential

4. Other Requirements

4.2 Legal, Regulatory and Organizational Requirements

These requirements apply the legal, ethical and policy constraints identified in the Phase 2 Legal and Regulatory Analysis [5], based on UAE legislation [6] to [10], and add the organizational requirements under which the prototype is developed. Data requirements (4.1) are written separately.

Most requirements below are verified by testing the system. A few (LEG-2, ORG-1 and ORG-4) concern the development process rather than system behavior, so they are verified by inspecting a document or record, in the same way as the organizational requirements described in the project description. Open questions that are not yet requirements, such as ownership of the codebase, are listed in Appendix C.

ID	Requirement	How it is measured	Use case	Priority
LEG-1	Data protection (UAE PDPL, Federal Decree-Law No. 45 of 2021 [6]): the system shall meet SEC-7 to SEC-11 (data minimization, consent, access and correction, human review of automated decisions, no personal data sent to AI services).	Compliance checklist reviewed at the end of system testing: every item passes.	UC-8, UC-9, UC-11, UC-13	Essential
LEG-2	A data protection impact assessment shall be drafted during the design phase, because the AI matching performs systematic automated evaluation (PDPL, Article 21), and shall be approved by Khalifa University before the system processes real university user data.	Inspection: the draft assessment is part of the design document; an approved assessment exists before any pilot with real accounts.	None (process requirement)	Essential
LEG-3	Until Khalifa University IT confirms otherwise (TBD-9), the system shall send no personal data to any service outside the UAE, since the PDPL regulates transfers of personal data abroad; the location where data is stored shall be documented and confirmed with KU IT before real data is used.	Inspection of all requests to external services during testing: 0 personal fields (same check as SEC-11). Inspection: the data storage location is documented.	UC-9, UC-10, UC-11, UC-12	Essential
LEG-4	Unauthorized access (UAE Cybercrime Law, Federal Decree-Law No. 34 of 2021 [8]): the system shall prevent and record unauthorized access through SEC-1 to SEC-6, SEC-12 and the audit log (FR-7.A1 to FR-7.A6).	Security tests of the listed requirements all pass.	UC-1 to UC-13	Essential

ID	Requirement	How it is measured	Use case	Priority
LEG-5	AI-generated content (UAE Copyright Law, Federal Decree-Law No. 38 of 2021 [7]): the terms of use shall state that system-generated reports and match explanations belong to Khalifa University, and the human review and approval of AI reports shall be recorded (FR-12.9, FR-12.10).	Inspection: the terms of use contain the statement. Test: 100% of published AI reports have a recorded approver.	UC-12	Essential
LEG-6	Procurement: the system shall present reuse options and avoided-purchase figures as decision support only. It shall not approve or block purchases; the decision to buy new or reuse stays with the department's budget holder and Procurement under university procedures [11] (BR-10).	Review: no system function approves or blocks a purchase; avoided-purchase figures carry a decision-support statement.	UC-9, UC-12	Essential
LEG-7	Hazardous and electronic waste shall be handled through Facilities Management or an approved recycling vendor, and the recycling or disposal certificate shall be kept with the asset record (FR-5.24, FR-5.25, FR-5.30).	Test: a hazardous asset cannot be closed without a handling party and certificate record.	UC-7	Essential
LEG-8	Indicators shall support reporting to the Khalifa University Sustainability Office, in line with its 'Recycle, Reduce & Reuse' program [12] (FR-13).	The Sustainability Officer can export the indicators they may see (SEC-5) for a semester.	UC-12	Desirable
LEG-9	Fairness (Phase 2 ethical considerations and risk R7): the matching criteria and their weights shall be documented and visible to Administrators, and AI ranking shall not use the requesting department's name or size as a factor (FR-8.5), so smaller departments are not disadvantaged.	Inspection: the criteria and weights are documented and visible to Administrators. The ranking itself is tested under QUA-13.	UC-9	Essential
ORG-1	The system shall follow Khalifa University IT data-governance and acceptable-use policy once obtained, and shall pass a KU IT review before handling real university accounts.	KU IT review recorded before any real pilot.	None (process requirement)	Future (required before a real pilot)
ORG-2	During the academic prototype, only synthetic or reviewed sample data shall be used; no real university personal data shall be loaded until ORG-1 and LEG-2 are met.	Review of the test data set before each demonstration: no real personal data.	None (process requirement)	Essential
ORG-3	The prototype shall use its own user accounts. Sign-in through the university's single sign-on shall be added only after a formal KU IT request and approval (TBD-3).	The prototype runs without any dependency on university sign-on.	UC-8, UC-13	Essential

ID	Requirement	How it is measured	Use case	Priority
ORG-4	Development shall be tracked in the course GitHub repository, with every team member committing their own work regularly during each phase.	Inspection of the commit history at the end of each phase: every member has commits for their own sections or code in that phase.	None (process requirement)	Essential

5. Use Cases

UC-6: Record Maintenance or Repair

Use case ID and name	UC-6: Record Maintenance or Repair
Description	Maintenance Staff record the inspection and repair or refurbishment work done on an asset with an open maintenance request, capture its cost, and update the asset's condition so it can return to service, or send it for end-of-life review if it cannot be repaired.
Primary actor (initiator)	Maintenance Staff
Secondary actors	Asset Coordinator and the user who reported the defect (notified of the outcome); Sustainability Officer (notified if the asset is flagged for end-of-life review)
Related requirements	FR-5.1 to FR-5.18; FR-7.N1 to FR-7.N5; FR-13.3; PER-3, PER-6; SAF-2, SAF-5, SAF-9; SEC-5, SEC-12, SEC-13; QUA-4, QUA-7; BR-7.
Preconditions	1. The user is signed in with the Maintenance Staff role. 2. A maintenance request for the asset is assigned to the user (FR-5.3) and is in the 'Reported', 'Inspection Requested', 'Under Inspection' or 'Repair in Progress' state.
Trigger	Maintenance Staff select an assigned maintenance request and choose 'Start work'.
Main flow	1. The system shows the asset's details, condition, maintenance history and safety notes, and sets the request to 'Under Inspection'. 2. Maintenance Staff record the inspection findings, the assessed condition and whether the asset is safe to use. 3. Maintenance Staff record the repair or refurbishment work, its cost and the estimated additional years of use (the request moves to 'Repair in Progress'). 4. Maintenance Staff enter the asset's updated condition and mark the request 'Completed'. 5. The system validates the data and saves it, updating the asset's condition, availability, life-cycle history and maintenance cost. 6. The system notifies the Asset Coordinator and the user who reported the defect.
Alternative flows	A1 (step 2) No repair needed: Maintenance Staff skip step 3 and complete the request after the inspection. A2 (step 2) Not repairable: Maintenance Staff close the request as 'Closed – Not Repairable'; the system flags the asset for end-of-life review and notifies the Asset Coordinator and the Sustainability Officer (FR-5.12), who may start UC-7. A3 (step 3) Work not finished: Maintenance Staff save progress; the request stays 'Repair in Progress' and the asset stays unavailable. The use case resumes later at step 3. A4 (step 2) Asset unsafe: Maintenance Staff mark the asset not safe to use; the system withdraws its listings and blocks matching, reservation and transfer (SAF-5).

Use case ID and name	UC-6: Record Maintenance or Repair
	A5 (step 5) Maintenance-cost threshold reached: the system flags the asset for end-of-life review and notifies the Asset Coordinator and the Sustainability Officer (FR-5.11).
Exception flows	E1 Missing or invalid data (for example a completion date before the start date, or a negative cost): the system identifies the invalid field and saves nothing. E2 The request is not assigned to the user, or the user lacks the Maintenance Staff role: the system denies access and records the attempt in the audit log. E3 The request was cancelled in the meantime (FR-5.13): the system informs Maintenance Staff and saves no changes. E4 The audit entry cannot be recorded: the system does not save the work and shows an error.
Postconditions	Success: the maintenance record and cost are saved; the asset's condition and availability are updated; life-cycle events, notifications and audit entries are recorded. Failure: the asset record is unchanged and the user sees the reason.
Rules	R1 Only Maintenance Staff or Administrators may record inspection or repair results. R2 An asset cannot be shown as available, matched, reserved or transferred while inspection or repair is in progress. R3 Costs are recorded in AED and cannot be negative. R4 Life-cycle events cannot be edited or deleted. R5 A cumulative maintenance cost of 50% or more of the replacement price triggers end-of-life review (default, TBD-7).

UC-7: Donate, Recycle or Dispose of an Asset

Use case ID and name	UC-7: Donate, Recycle or Dispose of an Asset
Description	A Sustainability Officer proposes an end-of-life action for an asset (donation, recycling, disassembly for usable components, retirement or disposal). A different Sustainability Officer records a safety review, an Administrator approves the proposal, and after handover the system records the outcome, closes the asset and updates the sustainability indicators. Disposal is treated as the last option. An Asset Coordinator can also start this use case (FR-5.20); the steps are the same.
Primary actor (initiator)	Sustainability Officer
Secondary actors	Second Sustainability Officer (safety reviewer); Administrator (approver, BR-8); Asset Coordinator (consulted and notified); Facilities Management or an approved recycling vendor (handling party for hazardous items); Donation Recipient.
Related requirements	FR-5.11, FR-5.12, FR-5.16 to FR-5.34; FR-7.N1 to FR-7.N5; FR-7.A4; FR-10.1 to FR-10.8; FR-13.3, FR-13.6; AI-1 to AI-10; PER-3 to PER-6; SAF-1 to SAF-4, SAF-6, SAF-7; SEC-5, SEC-10, SEC-12; QUA-4, QUA-7, QUA-11 to QUA-14; BR-4, BR-8, BR-9; LEG-7.
Preconditions	1. The user is signed in with the Sustainability Officer role. 2. The asset is registered and is not already in a final state (Donated, Recycled, Disassembled, Retired or Disposed).

Use case ID and name	UC-7: Donate, Recycle or Dispose of an Asset
Trigger	The Sustainability Officer selects 'Propose end-of-life action' on an asset, usually one flagged for end-of-life review (UC-6).
Main flow	1. The system shows the asset's details, condition, history, maintenance cost, open requests it may satisfy, and the AI recommendation with its confidence and explanation (if available). 2. The Sustainability Officer selects the action, enters the justification (for disposal, a reason for each alternative) and confirms or updates the hazard flag. 3. The system checks for blocking items, saves the proposal as 'Pending Safety Review' and notifies a different Sustainability Officer. 4. The second Sustainability Officer records the safety review; the system sets 'Pending End-of-Life Approval' and notifies the Administrator. 5. The Administrator approves the proposal; the system notifies the proposer and the Asset Coordinator. 6. After handover, the Sustainability Officer records the completion details (recipient or handling party, date, quantity, weight, and the handover document or certificate); the system sets the asset's final state, adds the life-cycle event, removes the asset from listings and updates the sustainability indicators.
Alternative flows	A1 (step 1) The asset matches an open request: the Sustainability Officer asks the Asset Coordinator to offer it for internal reuse instead (UC-3 to UC-5); no proposal is created. A2 (step 2) The chosen action differs from the AI recommendation: the system requires a reason and records the override. A3 (step 4) The asset is hazardous or electronic waste: the reviewer assigns Facilities Management or an approved vendor (FR-5.25), and at step 6 the handling record and certificate are required. A4 (step 4 or 5) The reviewer or the Administrator rejects the proposal: the system records the reason, returns the asset to its previous state and notifies the proposer. A5 (step 5) The AI recommendation being followed has Low confidence: the Administrator must record their own reason before approving (SAF-6). A6 (step 6) Retirement: the completion details are the date, reason and final location (FR-5.31).
Exception flows	E1 The asset has an open reservation, pending transfer or open maintenance request: the system rejects the proposal and lists the blocking items. E2 The justification is missing, or a disposal proposal is missing the reason for any alternative: the system does not submit the proposal. E3 The safety reviewer or approver is the proposer: the system refuses the action. E4 The AI recommendation is unavailable: the system shows 'Recommendation unavailable' and the Sustainability Officer continues without it. E5 Required completion details are missing (such as weight, or the certificate for a hazardous item): the system does not close the asset.
Postconditions	Success: the asset is in its final state and cannot be reserved or matched; the completion record, life-cycle event, notifications and audit entries are saved; the indicators include the action. Rejected or failed: the asset is in its previous state, and the proposal is kept with its reason.

Use case ID and name	UC-7: Donate, Recycle or Dispose of an Asset
Rules	R1 Disposal is the last option and needs a reason for rejecting each alternative: reuse, repair, donation and recycling. R2 The proposer cannot perform the safety review or approve their own proposal. R3 Hazardous items are never closed without a safety review, a handling party and a certificate. R4 AI recommendations are advisory only and never complete an action automatically. R5 Weight is recorded for every donation, recycling, disassembly and disposal (actual, category default or estimate, marked as such).

UC-12: Generate a Savings and Sustainability Report

Use case ID and name	UC-12: Generate a Savings and Sustainability Report
Description	A Finance Officer generates the savings and sustainability report (FR-12.2 type b) for a selected period. The system calculates the indicators with fixed formulas, obtains an AI-written summary of those figures, checks that the summary's figures match, and publishes the report only after the Finance Officer reviews and approves it.
Primary actor (initiator)	Finance Officer
Secondary actors	Sustainability Officer (optional reviewer); University Management (receives the published report).
Related requirements	FR-7.N1 to FR-7.N5; FR-7.R1 to FR-7.R8; FR-7.A4; FR-12.1 to FR-12.12; FR-13.1 to FR-13.13; AI-1 to AI-10; PER-4, PER-5; SAF-2; SEC-5, SEC-11; QUA-14; BR-9 to BR-11; LEG-3, LEG-5, LEG-6, LEG-8.
Preconditions	The user is signed in with the Finance Officer role.
Trigger	The Finance Officer selects 'Generate savings and sustainability report'.
Main flow	1. The Finance Officer selects the report period (for example a month) and the scope (campus or a department). 2. The system calculates the indicators for that period and scope (FR-13) and shows them with their data coverage, assumptions and missing data. 3. The Finance Officer requests the AI summary; the system sends only aggregated, non-personal figures to the AI service. 4. The system receives the draft, checks every figure against the calculated values, and shows it labeled 'AI-generated' with the model version, period, data sources and assumptions. 5. The Finance Officer edits the draft if needed and approves it; the system checks the figures in the edited version again. 6. The system stores the draft and the approved version, records the approval, publishes the report to authorized users (hiding financial figures from those without financial access) and notifies them.
Alternative flows	A1 (step 3) The Finance Officer needs only the figures: the system produces the standard report and indicators (FR-7, FR-13) without an AI summary, and the use case ends.

Use case ID and name	UC-12: Generate a Savings and Sustainability Report
	A2 (step 5) The Finance Officer is not satisfied: they regenerate the draft or discard it; a discarded draft is kept as 'Discarded' and nothing is published. A3 (step 5) The Finance Officer sends the draft to a Sustainability Officer for review before approving it.
Exception flows	E1 The AI service is unavailable or does not respond within the time limit in PER-5: the system informs the Finance Officer and offers the standard report and indicators instead. E2 There is no recorded activity in the selected period: the system says so and does not generate a summary. E3 A figure in the draft or edited version does not match the calculated value: the system highlights the statement and blocks approval until it is corrected or the draft is regenerated. E4 Some records lack a value an indicator needs and have no category default: the system excludes them and lists them under 'Missing data' (FR-13.8).
Postconditions	Success: the approved report is stored with its draft, model version, period and approver, and is available to authorized users; the audit log records the generation and approval. Failure: nothing is published; the standard report and indicators remain available.
Rules	R1 Only approved AI reports are shared with others. R2 Every figure comes from the fixed-formula calculations (FR-7, FR-13); the AI does not produce its own numbers. R3 Sustainability and savings figures are labeled as estimates, with their assumptions. R4 Financial values are in AED and visible only to roles allowed to see financial information (SEC-5). R5 Recommended actions are suggestions only and are never carried out automatically.

AI Contributions

Rashid Alzaabi

An AI assistant (Claude, Anthropic) was used to help organize the requirements in sections 2.5, 2.7 (reports and audit log), 2.12, 2.13, 3.2, 3.3 and 4.2 and the use cases UC-6, UC-7 and UC-12. It was also used to check them against the project description, and the commitments made in the Phase 2 feasibility study. All requirements, measurements and default values were reviewed and edited by the author, and values that still need confirmation from Khalifa University are listed as TBD in Appendix C.